

21	C	$I = nAvq$ $v = \frac{I}{nAq} = \frac{8.0}{8960 / (1 \times 10^{-25}) \times \pi (4.0 \times 10^{-3})^2 \times 1.6 \times 10^{-19}} = 1.1 \times 10^{-5} \text{ m s}^{-1}$ Distractors: 2.8×10^{-6}: Never convert diameter to radius.
22	D	When the switch is open, at balance position, there is no current passing through the unknown emf cell